

JEE MAIN REVISION MASTER: PROBABILITY 2026

5 QUESTIONS 8.33% WEIGHTAGE ↓16.7%

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 44 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Probability - Core Application	44 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Probability - Core Application (Very High Yield — 44 PYQs)

Core Principle: Key problem pattern in Kcet Probability.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Recent studies suggest that 12% of the world population is left handed. Depending on parents' hand usage, the chances of having left hande..." [Ans: N/A]
- [2026 Shift] Q#1 (MCQ): "Probability of at least one of the events A and B occur is 0.6 . If A and B occur simultaneously with probability 0.2 , then $P(\bar{A})$..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "The probability that a man and his wife live after 20 years are (1 / 4) and (1 / 3) respectively. The probability that neither..." [Ans: D]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Recent studies suggest that 12% of the world population is left handed. Depending on parents' hand usage, the chances of having left handed children are as follows: A: Both parents are left handed, chances of having left handed children =24% B: Both parents are right handed, chances of having left handed children =9% C: Father left handed and mother right handed, chances of having left handed children =17% D: Father right handed and mother left handed, chances of having left handed children =22% Given $P(A)=P(B)=P(C)=P(D)=1/4$ and L denotes child is left handed. What is the probability that $P(A|L)$?

- (A) (17 / 100)
- (B) (19 / 25)
- (C) (1 / 3)
- (D) (2 / 3)

Correct Answer: None

Solution Analysis: We need to find $P(A|L)$, which means: Probability that both parents are left handed, given that the child is left handed. Using Bayes' theorem, $P(A|L)=(P(A)P(L|A) / P(L))$ Here, $P(A)=P(B)=P(C)=P(D)=(1 / 4)$ Also, $P(L|A)=24\%=(24 / 100)$ Now find $P(L)$ using the total probability formula:

$P(L)=P(A)P(L|A)+P(B)P(L|B)+P(C)P(L|C)+P(D)P(L|D)$ Substitute the values: $P(L)=(1 / 4) \times (24 / 100)+(1 / 4) \times (9 / 100)+(1 / 4) \times (17 / 100)+(1 / 4) \times (22 / 100)$ Since $(1 / 4)$ is common, $P(L)=(1 / 4)((24+9+17+22 / 100))$ $P(L)$